Rockchip Linux DMABUF Developer Guide

ID: RK-KF-YF-457

Release Version: V1.0.0

Release Date: 2022-05-02

Security Level: □Top-Secret □Secret □Internal ■Public

DISCLAIMER

THIS DOCUMENT IS PROVIDED “AS IS”. ROCKCHIP ELECTRONICS CO., LTD.(“ROCKCHIP”)DOES NOT PROVIDE ANY WARRANTY OF ANY KIND, EXPRESSED, IMPLIED OR OTHERWISE, WITH RESPECT TO THE ACCURACY, RELIABILITY, COMPLETENESS,MERCHANTABILITY, FITNESS FOR ANY PARTICULAR PURPOSE OR NON-INFRINGEMENT OF ANY REPRESENTATION, INFORMATION AND CONTENT IN THIS DOCUMENT. THIS DOCUMENT IS FOR REFERENCE ONLY. THIS DOCUMENT MAY BE UPDATED OR CHANGED WITHOUT ANY NOTICE AT ANY TIME DUE TO THE UPGRADES OF THE PRODUCT OR ANY OTHER REASONS.

Trademark Statement

"Rockchip", "瑞芯微", "瑞芯" shall be Rockchip’s registered trademarks and owned by Rockchip. All the other trademarks or registered trademarks mentioned in this document shall be owned by their respective owners.

All rights reserved. ©2022. Rockchip Electronics Co., Ltd.

Beyond the scope of fair use, neither any entity nor individual shall extract, copy, or distribute this document in any form in whole or in part without the written approval of Rockchip.

Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian, PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

This document introduces a debugging method for DMABUF, which can effectively analyze and confirm issues such as DMABUF memory leaks, data output errors, and memory mapping failures.

Product Version

Chip Models	Kernel Version
RK3588	Linux-5.10

Target Audience

This document (this guide) is primarily intended for the following engineers:

Technical Support Engineers

Software Development Engineers

Revision History

Version	Author	Date	Description
V1.0.0	Xu Jianqun	2022-05-02	Initial version

Table of Contents

Rockchip Linux DMABUF Developer Guide

1. Introduction
2. DMABUF Debugging Modifications in Linux
   - 2.1 DMABUF_DEBUG
   - 2.2 DMABUF_PROCFS
   - 2.3 DMABUF_DEBUG_ADVANCED
3. QA
   - 3.1 Is it DMABUF Memory Leak Issue During Video Playback on RK3588?
   - 3.2 What is the Peak DMABUF Memory Usage During Video Playback on RK3588?

1. Introduction

DMABUF is a specific buffer structure for sharing memory between multimedia devices. Different devices can share memory across processes via the DMABUF file descriptor (fd), achieving Zero-Copy to improve performance and supporting asynchronous access to reduce module coupling.

2. DMABUF Debugging Modifications in Linux

2.1 DMABUF_DEBUG

Starting from Linux-5.10, DMABUF debugging is supported. To enable this feature, modify `arch/arm64/configs/rockchip_defconfig`.

```
CONFIG_DMABUF_DEBUG=y
```

When this macro is enabled, each DMABUF is assigned a default name in the format "pid+taskname", limited to 16 characters. For example:

```
416-allocator@4.
```

2.2 DMABUF_PROCFS

Starting from Linux 5.10, DMABUF procfs is supported. Enabling this feature requires modifying `arch/arm64/configs/rockchip_defconfig`.

```
CONFIG_RK_DMABUF_PROCFS=y
```

Enable this macro definition to create a rk_dmabuf subnode under the /proc/ directory. This subnode supports various debug nodes, such as

- dev # Display All Devices Attached to DMABUF
- sgt # Display the ScatterListTable of all DMABUFs and show their IOVA
- size # Display the total size of all DMABUFs
- peak # Display the peak total size of all DMABUFs

2.3 DMABUF_DEBUG_ADVANCED

This is a sub-feature of the DMABUF procfs interface, used to retrieve the physical memory addresses of DMABUFs. Enabling this feature requires modifying `arch/arm64/configs/rockchip_defconfig`.

```
CONFIG_DMABUF_DEBUG_ADVANCED=y
```

Enable this macro definition to ensure all DMABUFs are attached/mapped via the fake device, thereby retrieving their SGTs and assisting /proc/rk_dmabuf_sgt in fully displaying the IOVA of all DMABUFs.

3. QA

3.1 Is it DMABUF Memory Leak Issue During Video Playback on RK3588?

Enable the debug macro and enter the main interface.

```
console:/ # cat /proc/rk_dmabuf/size
Total: 65552 KiB
```

Play video, exit playback, delete history content, return to the main interface

```
console:/ # cat /proc/rk_dmabuf/size
Total: 83948 KiB
```

3.2 What is the Peak DMABUF Memory Usage During Video Playback on RK3588?

Check the total DMABUF memory size for video playback via the MPP debug node.

```
console:/ # cat /proc/mpp_service/sessions-summary
fdc38100.rkvdec-core Total: 699760 KiB
```

View peak DMABUF memory usage via the DMABUF debug node.

```
console:/ # cat /proc/rk_dmabuf/peak
Peak: 1544 MiB
```

Clear and observe again.

```
console:/ # echo 0 > /proc/rk_dmabuf/peak
```